SHORT PAPER

First version: October 2025 · This version: October 26, 2026

Do budgeting tools help low-income students save? An experiment with OBMEP scholarship holders and what self-reports can tell us

Gabriel Moreno Ribeiro

Fundação Estudar, Salvador, Bahia, Brazil · Advisor: Aaron Litvin, Ph.D., Harvard University

ABSTRACT

I randomly assigned 208 holders of the Junior Scientific Initiation Program (PIC) scholarship, which pays R$ 300 a month, to a control group or to one of three budgeting tools: a Google Sheets template, the Mobilis app, or both. For three months, students reported how much of the stipend they saved. In October, the control group reported saving 44.25 % of what was left after essential expenses, and the tool groups reported between 91.40 % and 101.07 %. Receiving any tool raised reported saving by 52.0 pp (95 % CI 44.2 to 59.8), and the estimate survives randomization inference, 108 alternative specifications and attrition bounds. The rest of the evidence is less encouraging. The three tools cannot be ranked. Students who never logged a single expense reported the same gains as those who did, which contradicts the attention channel of the model I propose. Reports show no persistence from one month to the next and are almost never round amounts. With synthetic data, I show that this experiment could not tell an effect that is the same for everyone from an effect that exists only in the reports. I conclude that the tools changed what students reported, but the data cannot say how much of that is real saving. I propose a follow-up study with bank balances and an attention placebo arm, whose analysis program has already been tested on synthetic data.

Keywords: saving; financial education; budgeting tools; present bias; field experiment; Brazil **JEL:** D14, D91, I22, O16

1 Introduction

I grew up watching families fall into debt through decisions made under pressure, and that made me want to understand how people with little money decide what to do with it. The Junior Scientific Initiation Program (PIC) is a good place to study the question. It pays R$ 300 a month to medalists of the Brazilian Mathematical Olympiad for Public Schools (OBMEP), and for some of them this may be the first regular income they manage on their own. This article asks whether a simple, free tool for keeping track of that money changes how much of it students save.

There are reasons to expect it would. When money is tight, attention narrows to whatever is due today (Mani et al., 2013; Mullainathan and Shafir, 2013). People put too much weight on the present (Laibson, 1997), and money that is not set aside for something tends to get spent (Thaler, 1999). A budgeting tool makes small purchases visible, like a snack on the way home from school, and puts a number on the saving goal. Reminders alone have raised saving in field experiments (Karlan et al., 2016). There are also reasons for doubt: financial education courses change behavior very little (Fernandes et al., 2014; Kaiser et al., 2022), and a tool only helps if it gets used.

This text revises and condenses the first version of the project, the October 2025 paper (Ribeiro, 2025a) and the December 2025 extended abstract (Ribeiro, 2025b). I went back to each student's data, redid the analysis with public code, wrote a model whose predictions the data can reject, and asked how far the reported numbers can be trusted. Of the 24 numerical claims in the first version, 13 match the data and 2 cannot be reproduced; one of those is the link between using the tool and saving, which comes up in Section 5.

2 The experiment

Of the 389 active PIC scholarship holders who were contacted, 208 returned the interest and consent form and make up the sample. All of them had a smartphone and internet access, and none had ever used a budgeting spreadsheet or app in a systematic way. Students were randomized within 7 school-grade strata into four groups of 52: control (G1), a Google Sheets template (G2), the Mobilis app (G3), and a hybrid group with both tools (G4). Randomizing within grade is like flipping a separate coin in each class: every group ends up with the same mix of grades.

Before the intervention, everyone was invited to a 60-minute online workshop on budgeting and saving goals, and the recording was shared afterwards. The common part was the same for all four groups; the treated students then spent 30 minutes in breakout rooms learning their tool. The spreadsheet had an entry sheet, a monthly summary and a dashboard with a "thermometer" for the saving goal. In the app, students set up categories and budgets and turned on weekly reminders to log their expenses. The intervention ran from August 1 to October 31, 2025.

At the end of each month, every student reported in an online form how much they had saved. The **outcome** is that amount as a percentage of baseline residual income, what is left of the stipend after the essential expenses measured at baseline:

$$S_{it} = 100 \times \frac{\text{saved by student } i \text{ in month } t}{R\$\,300 - e_i}, \tag{1}$$

where e_i is baseline essential spending (R$ 156.32 on average, leaving R$ 143.68 of residual income). Because the denominator is fixed, S_{it} can exceed 100. Treated students also sent a screenshot of their tool, which is where the number of expense logs per week comes from. 203 students finished the study and 5 dropped out.

I estimate the effect of each tool on October saving with a regression that controls for baseline saving S_{i0}, covariates X_i and stratum fixed effects $\mu_{s(i)}$:

$$S_{i3} = \alpha + \sum_{g=2}^{4} \beta_g T_{gi} + \rho\, S_{i0} + X_i'\gamma + \mu_{s(i)} + u_i\,, \tag{2}$$

where $T_{gi} = 1$ if the student was assigned to group g, with robust (HC2) standard errors. The study was not pre-registered, and the first version reports neither ethics committee approval nor trial registration. The first version specified the outcome, the ANOVA, the regression on group indicators and the regression on tool use; everything else, including the model in the next section, was added after data collection and should be read as exploratory.

3 A model of attention and reporting

Student i has residual income R_i, plans discretionary spending c and saves the rest. Some small purchases also happen without a decision, like a soda after class; I call their total m_i and assume the student notices only a fraction a_{it} of it, which I call **attention** (Gabaix, 2019). The unnoticed part leaks out of the budget:

$$s_{it} = R_i - c_{it} - (1 - a_{it})\, m_i + \varepsilon_{it}\,, \tag{3}$$

where ε_{it} is a shock in that month. With utility $\log c + \beta v\, s$, where v is the value of each real saved and $\beta < 1$ is **present bias**, planned spending is $c^* = 1/(\beta v)$. A tool works through two channels. Logging expenses raises attention, $a_{it} = a_{0i} + \kappa u_{it}$, with u_{it} the logs per week. Seeing the goal every week raises the value of saving from v to $v + \Delta v$ (salience). The effect is

$$\Delta s_{it} = \underbrace{\kappa\, u_{it}\, m_i}_{\text{attention}} + \underbrace{\frac{1}{\beta v} - \frac{1}{\beta(v + \Delta v)}}_{\text{salience}}\,. \tag{4}$$

The first version called the tool a “cognitive prosthesis”. In the model, the prosthesis is a lamp held over the holes in a leaking bucket: the more often students switch it on, the more holes they see, and those who never switch it on stay in the dark. Five predictions follow.

P1. Receiving a tool raises saving.

P2. Among treated students, saving rises with use ($\kappa m_i > 0$); those who never log gain only through salience and should look more like the control group.

P3. The effect in reais depends on R_i: scarcity predicts larger gains for tight budgets, larger leaks predict larger gains for looser ones.

P4. Students who already saved a lot have less to gain.

P5. Since β, m_i and a_{0i} belong to the student, saving persists from one month to the next.

The study does not observe s_{it}, only what the student wrote down. I model the report as

$$r_{it} = s_{it} + b\,T_i + \eta_{it}\,, \tag{5}$$

where b is a **reporting bias** shared by treated students (a wish to show progress to the person who trained them, for instance) and η_{it} is random error. The bias b goes straight into the difference in means and satisfies P1, but it creates no slope on use. The error η leaves the means alone and acts on persistence: if true saving has correlation ρ across months and $\lambda = \mathrm{Var}(s)/[\mathrm{Var}(s) + \sigma_\eta^2]$ is the **reliability** of the reports, then

$$\mathrm{corr}(r_{it}, r_{it'}) = \rho\,\lambda\,. \tag{6}$$

P2 and P5 are therefore the predictions that separate saving from reporting.

4 Results

Figure 1 shows what happened month by month. The tool groups start close to the control group, between 37.55 % and 45.89 % of residual income, and pull away as early as August. In October they report 91.40 %, 97.82 % and 101.07 %, while the control group stays at 44.25 %. In money, average October saving was R$ 65 in the control group and between R$ 123 and R$ 148 in the tool groups.

Table 1 reports the regressions. Without controls, the effects are the differences in means, 47.1, 53.6 and 56.8 pp; adding baseline saving, covariates and fixed effects barely moves them. Pooling the three tools, receiving any of them raised reported October saving by 52.0 pp (95 % CI 44.2 to 59.8), about R$ 71 a month, an increase of 117 % over the control mean. P1 holds.

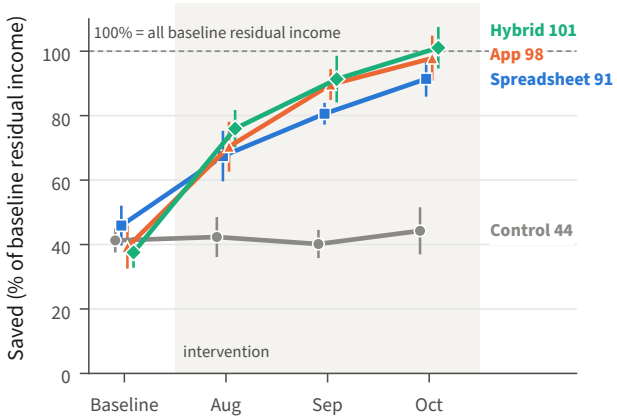


Figure 1. Reported saving as a percentage of baseline residual income, by group and month (mean and 95 % CI). The shaded band is the intervention period; the dashed line marks 100 %. Control in gray, spreadsheet in blue, app in orange and hybrid in green.

Source: study data, measured and estimated (`code/13_figures.py`).

The effect showed up early and kept growing. In an event study with student and month fixed effects, 55 % of the October effect was already there in August and 90 % in September,

Table 1. Effect of the tools on reported October saving. OLS with HC2 standard errors in parentheses. Columns (1) to (3): percentage of baseline residual income; column (4): reais, controlling for baseline saving in reais. $^{***}p < 0.01$.

	% of residual income saved			R$ saved
	(1)	(2)	(3)	(4)
Spreadsheet	47.15***	47.43***	46.86***	61.59***
	(4.67)	(4.69)	(4.70)	(6.93)
App	53.57***	53.45***	53.75***	75.59***
	(5.16)	(5.20)	(4.88)	(7.76)
Hybrid	56.82***	56.60***	55.68***	77.44***
	(4.97)	(5.01)	(4.89)	(7.35)
Baseline outcome	–	✓	✓	✓
Covariates, grade FE	–	–	✓	✓
Control mean	44.25	44.25	44.25	65.48
p: Spreadsheet = App = Hybrid	0.07	0.11	0.16	0.05
p: Hybrid = App	0.50	0.51	0.67	0.79
R^2	0.48	0.48	0.52	0.61
Observations	203	203	203	203

Source: study data, estimated (`code/03_main_effects.py`).

so growth was slowing down. Three months cannot tell where it would have leveled off, or whether it would last after the study.

The estimate survives every check I ran. None of 20,000 re-randomizations within strata produced a difference as large as the observed one (Young, 2019): the largest was 21.6 pp, against 52.5 pp in the data. Across 108 specifications (six versions of the outcome, three sets of controls, OLS or median regression, and three samples), the effect ranges from 1.2 to 3.8 standard deviations (SD) of the control group, all with $p < 0.001$ (Simonsohn et al., 2020). Lee (2009) bounds for attrition run from 51.4 to 53.8 pp, and assigning the worst possible value to every dropout still leaves an effect of 47.5 pp over the control group.

5 Which tool, and why

The hybrid group has the highest mean, followed by the app and then the spreadsheet, as in the first version. The data do not back that ranking. Hybrid and app differ by 3.2 pp ($p = 0.50$). Only hybrid versus spreadsheet reaches $p < 0.05$ (9.7 pp, $p = 0.03$), and it does not survive the Holm (1979) correction for six comparisons ($p = 0.08$). Nor can I say that the tools are equivalent: with a 10 pp margin, no pair passes the equivalence test. The experiment never had the power to rank them. With groups this size, the smallest difference between two groups it could detect with 80 % power is 13.5 pp; power for the observed gap between hybrid and app was 10 %, and detecting it would take about 855 students per group. The "synergy" between the two tools reported in the first version may exist, but this study offers no evidence of it.

The most direct test of the idea behind the study is P2: if the tool works by making expenses visible, students who used it more should have saved more. The first version reported a large positive slope. The data do not reproduce it (Figure 2). In percentage points, the slope of the change in saving on logs per week is –1.33 (95 % CI –3.19 to 0.54). The 25 treated students who never logged an expense (16 %) reported 96.6 % in October, the same as those who did (96.7 %, $p = 0.97$), and the most active third (4.8 logs per week) reported less than the least active third (92.1 % versus 101.9 %).

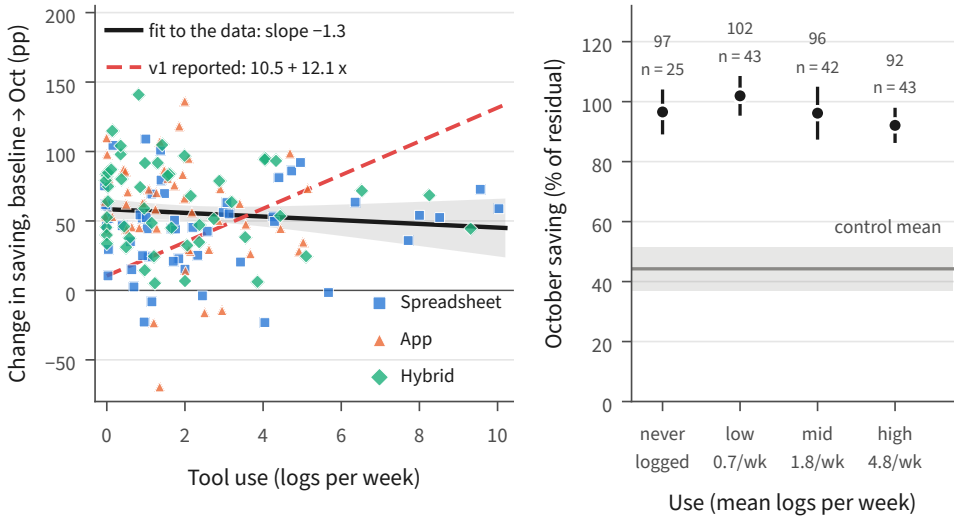


Figure 2. Tool use and reported saving among the 153 treated students. Left: change in saving from baseline to October against expense logs per week, with the OLS fit (95 % band) and the line reported in the first version (dashed). Right: October saving of students who never logged and of the usage terciles among those who did; the gray band is the control mean with its 95 % CI.

Source: study data, measured and estimated; first-version line, v1 (`code/13_figures.py`).

These comparisons are not causal, since each student chose how much to use the tool, and logs counted from screenshots are a rough measure of use. Still, in the model, students who never logged could only gain through salience, and they gained as much as everyone else. Either salience does all the work, or the effect runs through something all treated students shared: the training in the breakout room, the feeling of being part of an intervention, or the expectation of reporting progress to the person who trained them. More use did not go with more saving, so app engagement statistics would be a poor way to judge a program like this.

P3 and P4 find no support either. As a percentage of residual income, the effect changes by –7.3 pp per SD of residual income ($p = 0.04$), but that comes from the definition of the outcome, which divides by R_i; in reais, the gradient is R$ 6.5 per SD ($p = 0.27$). The interaction with baseline saving is not significant either ($p = 0.34$).

6 Putting the reports to the test

The experiment is internally valid for what it measured: randomization produced balanced groups, almost nobody dropped out, and the result does not hinge on analysis choices. The question is whether what it measured is saving. Treated students had been trained to

log their spending, knew the study was about saving, and reported their numbers directly to the researcher who ran the workshop. Experimenter demand effects tend to be small in online experiments (de Quidt et al., 2018), but being asked again and again can by itself change what people do and what they say (Zwane et al., 2011).

Persistence. P5 fails clearly. After removing group means, the correlation of a student's saving across consecutive months is 0.00 (95 % CI −0.08 to 0.08), and the intraclass correlation over the four months is −0.05. By Equation (6), this caps reliability: using the upper end of the interval, the share of report variance that is signal can be at most 0.27 if $\rho = 0.3$, 0.16 if $\rho = 0.5$ and 0.12 if $\rho = 0.7$ (Figure 3). Unless true saving is almost random from month to month, most of the variation in individual reports is noise.

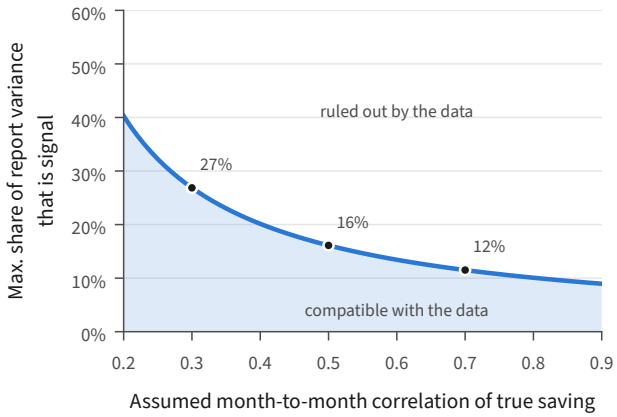


Figure 3. Largest share of the variance of monthly reports that can be signal, given the upper end of the 95 % CI of the month-to-month correlation, as a function of the assumed persistence of true saving (ρ, illustrative).

Source: study data, estimated (`code/13_figures.py`).

Saving more than the budget, and round numbers. In October, 43 % of treated students and 4 % of the control group reported saving more than 100 % of baseline residual income. One explanation would be reporting the running balance the tool displays instead of the month's saving; but then trajectories would climb month after month, and only 43 % of treated students have August < September < October, against 42 % if each student's months were shuffled at random. People who report money from memory usually round. Here, only 4 of the 208 essential-spending amounts are whole reais (1.9 %, against 1 % by chance), and saved amounts are whole reais in 2.2 % of cases and multiples of R$ 5 in 0.1 %, against 3 % and 0.6 % by chance. That includes the control group, which had no tool to copy an exact figure from. Precise amounts are not impossible, but they are one more reason to want to see the original forms.

A skeptical reader. An effect of 2.0 control-group SD is far larger than the typical effect of financial education on behavior, about a tenth of an SD (Kaiser et al., 2022). A skeptic who starts from a prior centered there (2.6 pp) and updates it with the 52.0 pp estimate ends up with a posterior mean of 17.7 pp if the prior SD is 0.1 SD, 38.8 pp with 0.25 SD and 47.9 pp

with 0.5 SD. That calculation, however, takes the reports at face value: a bias b that shifts every treated student goes through it unseen.

Synthetic worlds. To find out whether the experiment could separate the explanations, I generated reports for 203 students in six worlds, each with the study's group sizes, control means, average monthly effect and report SD (22 pp). The worlds combine three channels (in the attention world, each student's gain is proportional to use; in the equal-effect world, every treated student gains the same; in the reporting-only world, nobody saves more and treated students only report more) with two reliabilities (faithful reports, or reports in which only a share $\lambda = 0.10$ of the variance is signal). In all of them, true saving has a month-to-month correlation of 0.5, an illustrative value. In each world I analyzed 1,000 synthetic samples with the same calculations run on the real data (Figure 4).

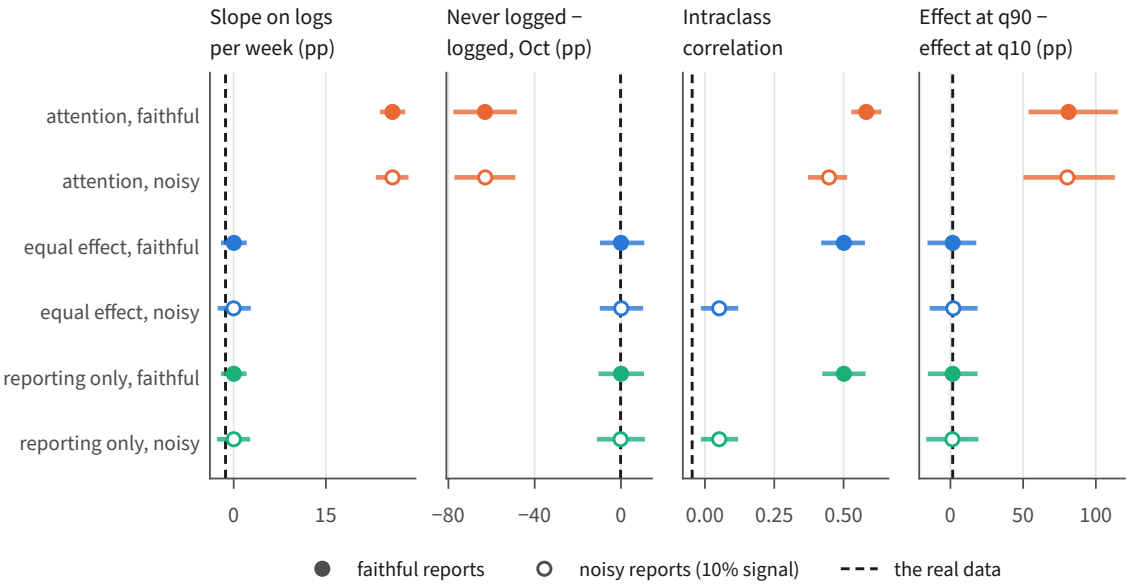


Figure 4. Four diagnostics in six synthetic worlds (mean and 95 % range over the 1,000 samples of each world; filled circle, faithful reports; open circle, noisy reports). Left to right: slope of the change in saving on logs per week; October saving of students who never logged minus that of students who did; intraclass correlation within groups; effect at the 90th percentile minus effect at the 10th. The dashed line is the value in the real data.

Source: simulated, synthetic data calibrated to the study data (`code/15_synthetic_worlds.py`); dashed line, study data, estimated.

The attention worlds sit far from the data on three diagnostics. There, the average slope is 25.8 pp per weekly log, saving of never-loggers minus loggers is –63.0 pp, and the effect at the 90th percentile exceeds the one at the 10th by 81.2 pp; in the data, the same quantities are –1.3 pp, the gap between 96.6 % and 96.7 %, and a quantile gap of 1.6 pp. The intraclass correlation in the data (–0.05) is far from the faithful worlds (0.50) and closer to the noisy ones (0.05), but still below their 95 % band (–0.01 to 0.11): the real reports look even noisier than those of the noisy world I simulated. The most important result is a negative one: the equal-effect and reporting-only worlds give the same numbers on all four diagnostics. With the data that exist, no calculation separates a real effect that is the same for everyone from an effect that lives only in the reports. That does not show the

effect is false. It shows that the evidence supports a narrower claim than the first version made: receiving a budgeting tool greatly increased the saving that students *reported*.

7 A follow-up study

What is missing is an outcome that does not depend on the student's own report, and a group that gets the same attention without the tool. I propose randomizing PIC scholarship holders, within grade strata, into three groups: control; an attention placebo, with the same workshop format, the same breakout room and weekly messages on a topic unrelated to money, such as study habits; and the simplest of the three tools tested here. The primary outcome would be the balance of a savings account opened for the study, or reported by the bank with the consent of the student and a guardian, at 3 and 12 months. The main test compares tool and placebo on that balance (H1); placebo versus control measures the effect of attention alone (H2); and the gap between what students report and the change in their balance, compared between tool and control, measures reporting bias (H3). The study would be registered before randomization and approved by a research ethics committee, as Brazilian rules require for research with people (Brasil. Conselho Nacional de Saúde, 2012).

I set the sample size by simulation, with 1,000 synthetic samples per scenario, a balance SD of R$ 55 and a correlation of 0.5 between starting and final balance, each sample analyzed by the same program that will be used on the real data (`code/analyze_followup.py`). For an effect of 0.3 SD, the power of H1 is 86 % with 150 students per group and 93 % with 200; for 0.2 SD, it reaches 81 % only with 300 per group.

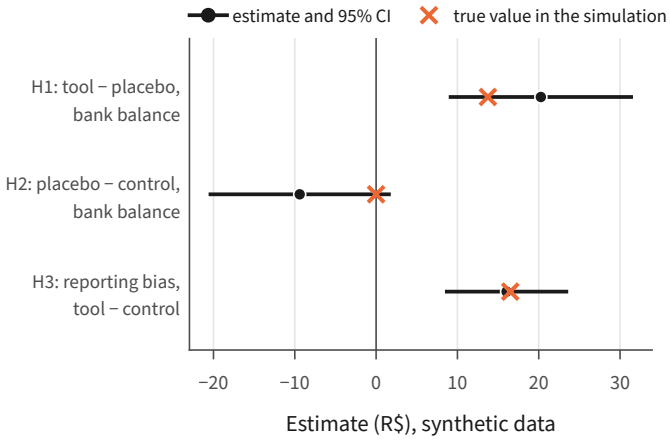


Figure 5. Dry run of the proposed study's analysis program on one synthetic data set (150 students per group). Dots and bars: estimates and 95 % CIs, in reais; crosses: the true values used to generate the data.

Source: simulated, synthetic data (`code/16_followup_proposal.py`).

The analysis program is already written, but it has only been run on synthetic data. In a dry run with 150 students per group, a true tool effect of 0.25 SD (R$ 14) and a reporting bias of 0.3 SD (R$ 16.5) in the tool group only, both illustrative values, the program estimated H1 at R$ 20.3 (95 % CI 9.2 to 31.4), H2 at R$ −9.4 (−20.4 to 1.6) and H3 at R$ 16.0 (8.7 to 23.4);

all three intervals contain the true value (Figure 5). The design separates the two things this experiment cannot.

8 Conclusion

Giving PIC scholarship holders a free spreadsheet or app and an hour-long workshop more than doubled the share of residual income they reported saving, from 44.25 % to between 91.40 % and 101.07 %. The result passes every statistical check I ran. What I cannot say is why it happened, or how much of it is real saving. The tools cannot be ranked, using the tool more did not go with saving more, and the reports do not behave like careful records of real behavior. The sample is OBMEP medalists, who are not typical students, and follow-up lasted three months. Since the intervention costs almost nothing, it is worth finding out whether it works, and the follow-up study, with bank balances and a placebo arm, was designed to answer that question.

Code and data availability. The code is in the `code/` folder of this project's repository, and every figure and table names the script that produced it. Every number in the text comes from `results/numbers.json`; synthetic data sets are in `data/simulation/`, and the first line of each file says they are not measurements.

References

Brasil. Conselho Nacional de Saúde (2012). Resolução nº 466, de 12 de dezembro de 2012. Diário Oficial da União, Section 1, Brasília, DF, no. 12, p. 59, 13 Jun. 2013. https://www.gov.br/conselho-nacional-de-saude/pt-br/atos-normativos/resolucoes/2012/resolucao-no-466.pdf. Resolution No. 466 of 12 December 2012, Brazilian National Health Council. Accessed 26 Oct. 2026.

de Quidt, Jonathan, Johannes Haushofer, and Christopher Roth (2018). Measuring and bounding experimenter demand. *American Economic Review*, 108(11):3266–3302. doi:10.1257/aer.20171330.

Fernandes, Daniel, John G. Lynch, Jr., and Richard G. Netemeyer (2014). Financial literacy, financial education, and downstream financial behaviors. *Management Science*, 60(8):1861–1883. doi:10.1287/mnsc.2013.1849.

Gabaix, Xavier (2019). Behavioral inattention. In Bernheim, B. Douglas, Stefano DellaVigna, and David Laibson, eds., *Handbook of Behavioral Economics: Foundations and Applications*, vol. 2, pp. 261–343. North-Holland. doi:10.1016/bs.hesbe.2018.11.001.

Holm, Sture (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.

Kaiser, Tim, Annamaria Lusardi, Lukas Menkhoff, and Carly Urban (2022). Financial education affects financial knowledge and downstream behaviors. *Journal of Financial Economics*, 145(2):255–272. doi:10.1016/j.jfineco.2021.09.022.

Karlan, Dean, Margaret McConnell, Sendhil Mullainathan, and Jonathan Zinman (2016). Getting to the top of mind: How reminders increase saving. *Management Science*, 62(12):3393–3411. doi:10.1287/mnsc.2015.2296.

Laibson, David (1997). Golden eggs and hyperbolic discounting. *Quarterly Journal of Economics*, 112(2):443–478. doi:10.1162/003355397555253.

Lee, David S. (2009). Training, wages, and sample selection: Estimating sharp bounds on treatment effects. *Review of Economic Studies*, 76(3):1071–1102. doi:10.1111/j.1467-937X.2009.00536.x.

Mani, Anandi, Sendhil Mullainathan, Eldar Shafir, and Jiaying Zhao (2013). Poverty impedes cognitive function. *Science*, 341(6149):976–980. doi:10.1126/science.1238041.

Mullainathan, Sendhil and Eldar Shafir (2013). *Scarcity: Why Having Too Little Means So Much*. Times Books, New York.

Ribeiro, Gabriel Moreno (October 2025a). The impact of financial organization technological tools on the savings capacity of scholarship students from the Scientific Initiation Program (PIC). Salvador: Fundação Estudar. Unpublished paper. Advisor: Aaron Litvin.

Ribeiro, Gabriel Moreno (December 2025b). The impact of financial organization technological tools on the savings capacity of scholarship students from the Scientific Initiation Program (PIC). Extended abstract (Behavioral Economics & Public Policy: Financial Tech Tools). Unpublished. Dated 27 Dec. 2025.

Simonsohn, Uri, Joseph P. Simmons, and Leif D. Nelson (2020). Specification curve analysis. *Nature Human Behaviour*, 4(11):1208–1214. doi:10.1038/s41562-020-0912-z.

Thaler, Richard H. (1999). Mental accounting matters. *Journal of Behavioral Decision Making*, 12(3): 183–206. doi:10.1002/(SICI)1099-0771(199909)12:3<183::AID-BDM318>3.0.CO;2-F.

Young, Alwyn (2019). Channeling Fisher: Randomization tests and the statistical insignificance of seemingly significant experimental results. *Quarterly Journal of Economics*, 134(2):557–598. doi:10.1093/qje/qjy029.

Zwane, Alix Peterson, Jonathan Zinman, Eric Van Dusen, William Pariente, Clair Null, Edward Miguel, Michael Kremer, Dean S. Karlan, Richard Hornbeck, Xavier Giné, Esther Duflo, Florencia Devoto, Bruno Crepon, and Abhijit Banerjee (2011). Being surveyed can change later behavior and related parameter estimates. *Proceedings of the National Academy of Sciences*, 108 (5):1821–1826. doi:10.1073/pnas.1000776108.